A Decentralized Voting Mechanism: Engaging ERC-20 token holders in decision-making

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***Abstract*—This paper presents a Blockchain-based voting mechanism that has the potential to disrupt current voting systems and fuel the development of decentralized governance. More specifically, the paper presents a study on the design and deployment of a *Solidity* smart contract that can interact with any Ethereum-based token (ERC-20) in order to help decentralized organizations to run public voting campaigns while at the same time engaging tokens holders in decision making. The use case scenarios outlined in the paper demonstrate with success the effectiveness of the proposed approach.**

***Index Terms*—voting mechanism, blockchain, smart-contract, decentralized governance**

I. Introduction

Blockchain networks firstly introduced by Satoshi Nakamoto [1] are considered to be one of the most vital technologies of our era [2] as it enables users to achieve secure transactions in a transparent, and trustworthy manner utilizing a peer-to-peer (p2p) network. In fact, blockchain is a distributed public ledger that entails numerous transactions in the form of blocks that are propagated into a network for validation. Once the blocks are validated and verified by the network peers (utilizing a consensus algorithm [3]) they are linked together expanding the depth of the network's data structure. A blockchain-based network is decentralized in nature as it enables parties to transact in a secure manner without the need of any intermediary. Among its unique characteristics, blockchain networks offer *transparency*; as all transactions are recorded on a public ledger, and *traceability*; as all blocks are linked together in reverse order. Such a data structure enables users to track the transactions recorded in previous blocks.

There has been an explosion in the implementation of various blockchain-based ecosystems in an attempt to evolve the technology and expand its capabilities in terms of interoperability and execution of machine code. Recent research from the community is concerned with improving or adapting consensus algorithms from distributed systems to blockchain networks [3] and expanding the use of the technology in other application domains as an attempt to further exploit the capabilities of the technology. Ethereum was introduced as a decentralized platform that enables the execution of smart contracts for the next generation of Web applications, along with its native cryptocurrency known as *Ether* used to incentivize the operations performed on the platform [4]. It was a first realistic attempt that enabled users to develop and deploy machine readable code (i.e., smart contracts) on a blockchain network. This postulated the opportunity to organizations to design trustful decentralized applications (dApps) that could run on a decentralized network without being controlled by a central authority [2], [5], as well as, the creation of Ethereum-based tokens that could run on the Ethereum network (e.g., ERC-20 [6]).

Token tracker, a service that tracks the status of ERC-20 smart contracts, provided by Etherscan[1] reports the existence of 246,064 token contracts. In general, ERC-20 is a technical standard that is being used for creating Ethereum-based tokens on the Ethereum blockchain. In most cases, these tokens have been used by various projects to represent a range of digital assets (e.g., digital tangible objects or utility tokens) which could be sold or exchanged with other tokens. In this paper, we investigate an alternative usage of ERC-20 tokens as a democratic, decentralized mechanism for voting, while at the same time enabling holders to participate in the decision-making of a Decentralized Autonomous Organization (DAO) to automate governance [7]. This algorithmic and democratized dimension of governance aligns with the transparency principle that blockchain-based systems represent. More specifically, the study presented in subsequent sections presents a smart contract implementation that enables any ERC-20 token implementation to participate in the organization and execution of voting (or execution of polls), thus engaging ERC-20 tokens holders in the decision making of a DAO.

The following questions motivated this study. *RQ1. How to engage users in the decision making of an organization?*, and *RQ2. How to convince companies to utilize an existing cryptocurrency as a payment method?*. The benefits of the proposed decentralized voting mechanism are summarized as follows. Firstly, users are becoming first-class citizens to the decision making of an organization by participating to various polls in a democratic way. Example cases of such decisions include: "a shop/website or other business where they want to spend their holdings", "the next exchange where a

[1]https://etherscan.io/tokens, Accessed 09 Mar. 2020

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cryptocurrency will be listed", "a place where an event can be hosted". On the other hand, organizations can collect valuable data that could be used for business intelligence before any formal decision is being taken. Lastly, organizations can reveal real market metrics which can be used to decide on whether a business should enable the acceptance of their cryptocurrency as a new payment method. Some of these metrics include, but are not limited to: "the number of unique users that voted for a specific business", "the amount of tokens that token holders can spend in a certain business".

The reminder of this paper is structured as follows. Section II presents a literature review on blockchain and smart contracts. Section III outlines the main characteristics of the proposed smart contract implementation along with a discussion on the various use-case scenarios, and Section III-C concludes the paper.

II. Blockchain and Smart Contracts

In 1997, Nick Szabo [8] firstly introduced the idea of smart contracts. Smart contracts can be described as self-executable computer programs that contain a set of rules which outline the terms of an agreement between interested parties. These rules can be automatically activated when certain conditions are met according to a certain logic that is embedded in a smart contract. The idea of smart contracts removes entirely the need of any trusted third-parties as you do not require an intermediate agent to verify a transaction. The Ethereum blockchain [4] introduced a pragmatic approach towards the execution of smart contracts over a blockchain-based environment that led to the development of decentralized applications. Ethereum clients (e.g., `geth`[2]) are responsible for executing the bytecode on the Ethereum Virtual Machine (EVM) that is associated with a contract account and only activated by an Externally Owned Account (EOA)[3] Interactions with smart contracts are triggered by transactions that are broadcasted to the network through smart contract function invocations either for reading or updating the EVM state. Transactions that happen on the public address of a smart contract are recorded on a public distributed ledger thus they cannot be altered or questioned making them transparent, secure and traceable.

The advantages of smart contracts led rapidly to the development of numerous decentralized applications in various domains of application. Chen [9] was one of the first works that outlined how Blockchain tokens can democratize entrepreneurship principles by offering to entrepreneurs innovative ways on raising funds by developing and deploying decentralized applications. Christodoulou *et al.* [10] presents the implementation of a smart contract tailored for the logistics and supply chain industry, providing a real-world example on how blockchains can be utilized to server the industry. Corsi et al. [11] introduces `TickEth`, an Ethereum-based smart contract that tackles the challenge of validating the authenticity of online tickets. Bogner *et al.* [12] presents a dApp developed on Ethereum that enables users to register and rent devices without the involvement of a Trusted Third Party (TTP) or the disclosure of any personal information. In addition, blockchain-based networks can be combined with IoT devices. Authors in [13] introduce a decentralized IoT application that makes use of the technology for serving a car park. In brief, `ParkChain` leverages image processing techniques, IoT devices, and blockchain smart contracts to track the time a registered vehicle accessed a car park. Finally, [14] presents a different use of blockchain as a protocol that facilitates low p2p exchange of ERC-20 tokens on the Ethereum blockchain namely the `0x` protocol. The 0x protocol serves as an open standard and common building block, driving interoperability among decentralized applications that incorporate token exchange functionality.

As it can be observed from the aforementioned decentralized applications many applications have been emerged to support different industrial domains or scientific fields, however, there is limited work on engaging ERC-20 token holders as a decentralized voting mechanism for enabling a decentralized form of governance for organizations. The proposed work inherits the capabilities of smart contracts and presents an opportunity for engaging users from open communities with the governance processes of a decentralized organization.

III. Voting Mechanism

A. Introduction

This paper presents the usage and implementation of an Ethereum-based smart contract that can be used as a voting mechanism. The implementation of the smart contract allows the utilization of any already deployed ERC-20 token to be used as part of the voting process, providing the capability to various organizations to create polls or other voting activities. Our motivation for this work is to investigate an alternative usage of ERC-20 tokens for holders that would like to use them as part of the decision making of some organization. The proposed implementation of the smart contract is developed using *Solidity* and can be executed on any blockchain network that uses the EVM. Fig. 1 presents a schematic of the proposed workflow that is embedded as part of the implementation. The reader is referred to the following `GitHub` repository for a complete access to the source code that implements the voting mechanism that can be used with any existing or newly deployed ERC-20 token *(Source code: https://github.com/panklitos/VotingMechanism)*.

B. How it works

The workflow of the proposed voting mechanism (depicted in Fig. 1) comprises of the following steps.

1) The contract owner (considered in this scenario as the Administrator of the organization) deploys the voting mechanism as encapsulated in the smart contract – on the Ethereum blockchain using as a variable to the

[2]https://geth.ethereum.org/

[3]*Externally Owned Accounts* are controlled by private keys as opposed to *Contract Accounts* that are controlled by private keys.

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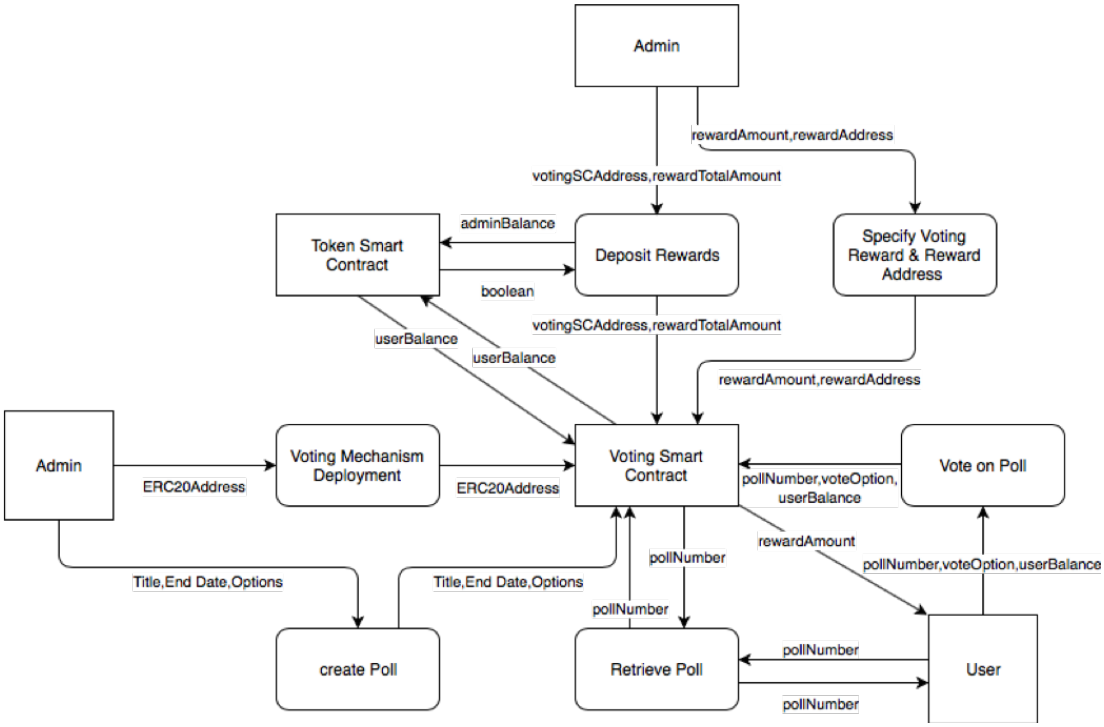

Fig. 1. A schematic representation of the proposed workflow implemented as an Ethereum-based smart contract.

constructor the contract address of any existing ERC-20 token contract.

2) Administrator[4] assigns tokens to the deployed voting mechanism contract; to be used as rewarding tokens to holders that voted (i.e., the *voters*).
3) Administrator sets the reward amount per vote and the *rewarding address*. Note that the *rewarding address* should be the same address as the one used by the user who deposited the total reward amount as part of *Step 2*.
4) At this step, the Administrator can create various polls by defining the following parameters: *poll-title, end-date*, and a *list* of the available options that token holders can vote for.
5) After the creation of a poll, a user that owns at least one token can vote (Note that each user can only vote once on each active poll).
6) To incentivize users to participate to polls, users are rewarded for successfully casting their vote with the specified ERC-20 token.

Note that all the data that are generated during the voting process are broadcasted as transactions to the network, recorded and verified by the consensus algorithm of the network. Lastly, results from the polls are publicly available and can be audited by interested parties.

[4]Admin is considered the poll creator

C. Real-world Demonstration

This section presents a preliminary experimental scenario that has a twofold purpose: (i) to test and validate the proposed approach, and (ii) to explore whether the users are incentivised to participate to decentralized polls.

Experimental Setup. Below we describe the experimental setup of the experimental scenario.

- **Bootstrapping:** Voting tokens with a total supply of 250M tokens were minted with the deployment of a smart contract named "MyAdvancedToken", following the ERC-20 token standard. The deployment resulted to the following contract address: *0x692a70d2e424a56d2c6c27aa97d1a86395877b3a* on the Ropsten testnet. Then, a random distribution of different amounts of tokens occurred to 200 Ethereum EOA addresses.
- **Voting Mechanism:** A smart contract that wraps the voting mechanism (as depicted in Fig. 1) was deployed invoking the constructor of the contract with the contract address of the "MyAdvancedToken" contract. The deployment of the voting mechanism contractresulted to the following contract address: *0xbbf289d846208c16edc8474705c748aff07732db*.

Based on the initial setup, as described in Section III, the Administrator with the ETH address *0x14723a09acff6d2a60dcdf7aa4aff308fddc160c* assigns to the VotingMechanism smart contract address a number of tokens (e.g., 1000000 tokens) to be used as reward tokens to voters.

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In addition, the Administrator sets the reward amount per vote to 1000 tokens.

Experimental Scenario. To demonstrate the effectiveness of the proposed voting mechanism two polls were created and then token holders were asked to cast their vote on them. Table I presents the average costs in ETH and USD for creating a poll or voting on a poll.

TABLE I
AVERAGE TRANSACTION FEES

	ETH	USD[5]
Create a Poll Function	0.000299	$0.0658
Vote on a Poll Function	0.000131	$0.02882

For the first generated poll token holders were asked the following question: "Where to host the annual meeting of the company?" To generate such a poll the following arguments are used: *title:* Where to host the annual meeting of the company?; *end-date:* 18/07/2019 (Unix epoch time 1563618750), and the list of options to choose from: {Athens, London, Amsterdam, Tel Aviv}.

As shown in Fig. 2 a total of 133 token holders participated into the poll (66.5% of token holders) and the most popular choice was Athens with 57% followed by Tel Aviv with 23.3%. In addition, 133000 tokens were distributed as a reward to participants.

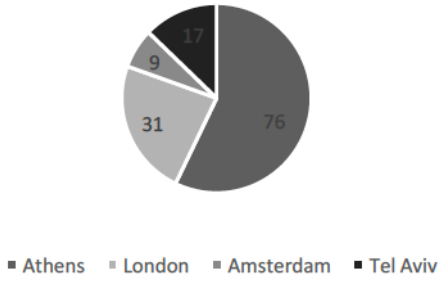

Fig. 2. Results of the annual meeting poll.

During the second poll token holders were asked the following question: "In which stores you may spend your tokens? To generate such a poll the following arguments are used: *title:* In which stores you may spend your tokens?; *end-date:* 23/07/2019 (Unix epoch time 1563877950), and the list of options to choose from: {Cafes/Restaurants, Bookshops, Clothing, Petrol Stations}. Fig. 3 presents the results of the second poll. A total of 101 token holders participated into the second poll (50.5% of token holders). The most popular choice was "Cafes/Restaurants" with 59.4% followed by "Clothing" with 18.8%. In addition, 101000 tokens were distributed as a reward to participants.

The aforementioned scenarios presented the opportunities of a decentralized voting mechanism smart contract to be used for online polls. Results showed that more than 50%

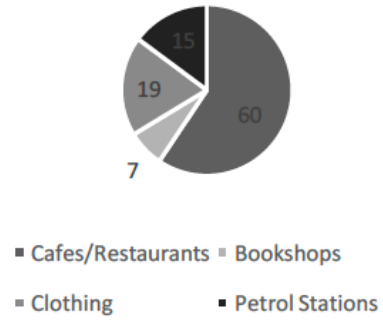

Fig. 3. Results of the annual meeting poll.

of token holders were interested in participating into polls. In addition, it is also observed that the participation rate depends on the question asked. Using the proposed voting mechanism organizations have the opportunity to engage with their decentralized communities and token holders involving them to the decision making. This could also be used as a tool for measuring the pulse of token holder's before proceeding with any final decision making concerning the organization.

D. Conclusions

Although there are plenty of decentralized applications alternative to the financial sector, there is limited work on engaging ERC-20 token holders as a decentralized voting scheme for enabling a decentralized form of governance for organizations. This study further supports the utility of Ethereum-based tokens by developing a smart contract that wraps a voting mechanism which can connect with any existing ERC-20 token contract. The proposed smart contract allows an organization to create decentralized polls engaging token holders in decision making. The benefits of the proposed decentralized voting mechanism are summarized as follows. Firstly, users are becoming first-class citizens to the decision making of an organization by participating to various polls in a democratic way. On the other hand, organizations can collect valuable data that could be used for business intelligence before any formal decision is being taken. Lastly, organizations can reveal real market metrics which can be used to improve the governance of a decentralized organization driven by a community of token holders. A set of experimental scenarios is used to explore the opportunities of a decentralized voting mechanism to be used for creating online polls. Results showed that more than 50% of token holders were incentivized to participate into decentralized polls utilizing their ERC-20 tokens.

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